Jinlin Lai

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Positions

Basis AI Institute *New York, NY*
SUMMER RESEARCH INTERN *June 2025 - August 2025*

Flatiron Institute, Simons Foundation *New York, NY*
SUMMER PRE-DOCTORAL RESEARCHER *May 2024 - August 2024*

Dolby Laboratories Inc. *Sunnyvale, CA*
ATG IMAGING RESEARCH INTERN *June 2023 - August 2023*

Education

University of Massachusetts Amherst *Amherst, MA, United States*
PH.D. IN COMPUTER SCIENCE *Aug 2020 - June 2026*

- Advisor: Dr. Daniel R. Sheldon
- Area: Probabilistic machine learning, computational statistics and Bayesian inference

University of Massachusetts Amherst *Amherst, MA, United States*
M.S. IN COMPUTER SCIENCE *Aug 2020 - Feb 2024*

Tsinghua University *Beijing, China*
B.ENG. OF COMPUTER SCIENCE AND TECHNOLOGY *Aug 2016 - June 2020*

- Minor in Finance and Entrepreneurship

Publications

PREPRINT

Jinlin Lai, Charles Margossian, Daniel Sheldon. (2026). Corrected Integrated Laplace Approximation for Bayesian Inference in Latent Gaussian Models. arXiv preprint arXiv:2605.20345. [link]

CONFERENCE

Max Hamilton, **Jinlin Lai**, Daniel Sheldon, Subhransu Maji. (2026). Scalable Model-Assisted Multi-Target Estimation in Large Image Collections. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence* (UAI 2026). To appear.

Jinlin Lai, Antonio Ricardo Linero, Yuling Yao. (2026). Predictive Variational Inference: Learn the Predictively Optimal Posterior Distribution. In *Proceedings of the 43rd International Conference on Machine Learning* (ICML), Seoul, South Korea. PMLR 306, 2026. To appear. [link]

Max Hamilton*, **Jinlin Lai***, Wenlong Zhao, Subhransu Maji†, Daniel Sheldon†. (2025). Active Measurement: Efficient Estimation at Scale. In *Proceedings of the 39th Conference on Neural Information Processing Systems* (NeurIPS). [link]

Jinlin Lai, Justin Domke, Daniel Sheldon. (2024). Hamiltonian Monte Carlo Inference of Marginalized Linear Mixed-Effects Models. In *Proceedings of the 38th Conference on Neural Information Processing Systems* (NeurIPS), Vancouver, Canada. [link]

Jinlin Lai, Anustup Choudhury, Guan-Ming Su. (2024). Outdoor Scene Relighting with Diffusion Models. In *Proceedings of the 27th International Conference on Pattern Recognition* (ICPR), Kolkata, India. [link]

Jinlin Lai, Javier Burroni, Hui Guan, Daniel Sheldon. (2023). Automatically Marginalized MCMC in Probabilistic Programming. In *Proceedings of the 40th International Conference on Machine Learning* (ICML), Honolulu, Hawaii, USA. PMLR 202, 2023. [link]

Jinlin Lai, Justin Domke, Daniel Sheldon. (2022). Variational Marginal Particle Filters. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics* (AISTATS) 2022, Valencia, Spain. PMLR: Volume 151. [link]

Haowen Xu, Wenxiao Chen, **Jinlin Lai**, Zhihan Li, Youjian Zhao, Dan Pei. 2020. Shallow VAEs with RealNVP Prior can Perform as Well as Deep Hierarchical VAEs. ICONIP.

Workshop

Aditya Parashar*, Aditya Vikram Singh*, Avinash Amballa*, **Jinlin Lai**, and Benjamin Rozonoyer. (2024). ”Quasi-random Multi-Sample Inference for Large Language Models.” In Frontiers in Probabilistic Inference: Learning meets Sampling. [link]

Jinlin Lai, Daniel Sheldon. 2022. Automatic Inference with Pseudo-Marginal Hamiltonian Monte Carlo. ICML workshop Beyond Bayes: Paths Towards Universal Reasoning Systems.

Jinlin Lai, Lixin Zou, Jiaxing Song. 2020. Optimal Mixture Weights for Off-Policy Evaluation with Multiple Behavior Policies. Offline Reinforcement Learning Workshop at Neural Information Processing Systems.

Services

Served as a reviewer for

- ICML 2022, 2025, 2026
- AISTATS 2023, 2024, 2025, 2026
- AABI (ProbML) 2023, 2024, 2026
- NeurIPS 2024, 2025, 2026
- ICLR 2025
- ACM SIGPLAN SLE 2025
- TMLR

Teaching Experience

Fall 2025	**Algorithms for Data Science**, Teaching Assistant	*University of Massachusetts Amherst*
Spring 2025	**Introduction to Computation**, Teaching Assistant	*University of Massachusetts Amherst*
Fall 2024	**Advanced Algorithms**, Teaching Assistant	*University of Massachusetts Amherst*
Spring 2024	**Probabilistic Graphical Models**, Teaching Assistant	*University of Massachusetts Amherst*
Spring 2023	**Probabilistic Graphical Models**, Teaching Assistant	*University of Massachusetts Amherst*
Spring 2022	**Probabilistic Graphical Models**, Teaching Assistant	*University of Massachusetts Amherst*
Summer 2019	**Algorithms for High School Olympics**, Lecturer	*Nanchang, Jiangxi Province, China*
Summer 2018	**Algorithms for High School Olympics**, Lecturer	*Ganzhou, Jiangxi Province, China*
Summer 2017	**Algorithms for High School Olympics**, Lecturer	*Ganzhou, Jiangxi Province, China*

Talks

Jinlin Lai. 2023. Automatically Marginalized MCMC in Probabilistic Programming. Contributed talk in *the 5th Symposium on Advances in Approximate Bayesian Inference*.